Immunology and Inflammation

Purging viral latency by a bifunctional HSV-vectored therapeutic vaccine in chronically SIV-infected macaques

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Abstract

The persistence of latent viral reservoirs remains the major obstacle to eradicating human immunodeficiency virus (HIV). We herein reported that recombinant herpes simplex virus type I (HSV-1) with ICP34.5 deletion could more effectively reactivate HIV latency than its wild-type counterpart. Mechanistically, HSV-ΔICP34.5 promoted the phosphorylation of HSF1 by decreasing the recruitment of protein phosphatase 1 (PP1α), thus effectively binding to the HIV LTR to reactivate the latent reservoirs. In addition, HSV-ΔICP34.5 enhanced the phosphorylation of IKKα/β through the degradation of IκBα, leading to p65 accumulation in the nucleus to elicit NF-κB pathway-dependent reactivation of HIV latency. Then, we constructed the recombinant HSV-ΔICP34.5 expressing simian immunodeficiency virus (SIV) env, gag, or the fusion antigen sPD1-SIVgag as an HIV therapeutic vaccine, aiming to achieve a functional cure by simultaneously reactivating viral latency and eliciting antigen-specific immune responses. Results showed that these constructs effectively elicited SIV-specific immune responses, reactivated SIV latency, and delayed viral rebound after the interruption of antiretroviral therapy (ART) in chronically SIV-infected rhesus macaques. Collectively, these findings provide insights into the rational design of HSV-vectored therapeutic strategies for pursuing an HIV functional cure.

eLife assessment

In this **useful** study, the authors tested a novel approach to eradicate the HIV reservoir by constructing a herpes simplex virus (HSV)-based therapeutic vaccine. The approach was tested in experimental infections of chronically SIV-infected, antiretroviral therapy (ART)-treated macaques with extent of rebound after ART interruption as a measure of the size of the HIV reservoir. While mean viremia at rebound was lower in the HSV vaccine-treated group, the evidence presented appear to be be **incomplete** because the group size was small and the viral load at rebound was highly variable.

Introduction

The epidemic of acquired immunodeficiency syndrome (AIDS), caused by human immunodeficiency virus type I (HIV-1), is still a huge challenge for global public health, with approximately 39 million people living with HIV-1 as of 2022. To date, there is neither a curable drug nor a prophylactic vaccine for clinical use against HIV-1 infections. Antiretroviral therapy (ART) can effectively control HIV-1 replication to an undetectable level, but the termination of ART usually results in prompt viral rebound from latent viral reservoirs (Archin et al., 2014, Looker et al., 2017, Calistri et al., 2003, Heng et al., 1994). Thus, it is of great priority to explore novel strategies for curing HIV latency. The "Shock and Kill" strategy is considered a promising approach for purging HIV-1 reservoirs, involving the activation of latently infected cells to express viral products (Shock), followed by viral cytopathic effects or specific cytolytic T lymphocytes (CTLs) to eliminate the activated cells (Kill) (Wu et al., 2022, Yang et al., 2019, Kim et al., 2018). Numerous latency-reversing agents (LRAs) (Yang et al., 2019, Wu et al., 2022), including methylation inhibitors, histone deacetylase (HADC) inhibitors (Archin et al., 2017, Lehrman et al., 2005), and bromodomain and extra terminal domain (BET) protein inhibitors (Li et al., 2013, Bisgrove et al., 2007), have been identified to reactivate latent HIV-1 in preclinical studies, but there is no ideal LRA available for clinical patients yet.

Herpes simplex virus (HSV), a human herpesvirus, features a 152-kb double-stranded DNA genome encoding over 80 proteins (Poh, 2016). Owing to its distinctive genetic background, high capacity, broad tropism, thermostability, and excellent safety profile, modified HSV variants have found extensive applications in gene therapy and oncolytic virotherapy. For example, talimogene laherparepvec (T-VEC), an HSV-1 variant with ICP34.5 deletion and GM-CSF insertion, received FDA approval in 2015 for treating malignant melanoma, showcasing notable safety and efficacy in clinical practice. Recombinant HSV-based constructs have also emerged as efficacious gene delivery vectors against infectious diseases. Early studies indicated that prophylactic vaccines based on HSV, expressing simian immunodeficiency virus (SIV) antigens, could elicit robust antigen-specific CTL responses in mice and monkeys, providing enduring and partial protection against pathogenic SIVmac239 challenges (Kaur et al., 2007, Murphy et al., 2000). Moreover, increasing data suggest the crucial role of HIV-specific CTL in controlling viral replication and eliminating potential HIV reservoirs (Collins et al., 2020, Leitman et al., 2017). Significantly, epidemiological research suggests a synergistic effect between HSV and HIV infections, with HSV infection in HIV patients being associated with increased HIV-1 viral load and disease progression (Looker et al., 2017, Calistri et al., 2003, Heng et al., 1994). Some studies have further unveiled the ability of HSV to activate HIV latent reservoirs (Amici et al., 2004, Amici et al., 2001, Pierce et al., 2023). Given the potential of HSV to simultaneously induce antigen-specific immune responses and reactivate latent viral reservoirs, we propose a proof-of-concept strategy to achieve an HIV functional cure using a modified bifunctional HSV-vectored therapeutic vaccine.

Results

The modified HSV-ΔICP34.5-based constructs reactivated HIV latency more efficiently than wild-type HSV counterparts

The J-Lat 10.6 cell line, originating from Jurkat T cells with latent HIV-1 provirus, was infected with wild-type HSV-1 17 strain (HSV-wt) at different multiplicities of infection (MOI) to assess its capability to reactivate HIV latency. Flow cytometry analysis showed that green fluorescent protein (GFP) expression was increased in a dose-dependent manner (***Figure 1A***), and the mRNA levels of HIV-1 LTR, Tat, Gag, Vif, and Vpr were also significantly increased in response to HSV stimulation (***Figure 1B***), demonstrating that the latent HIV provirus can be reactivated to a certain extent by wild-type HSV. In addition, different HSV-1 strains, including HSV-1 Mckrae and HSV-1 F strain, can also reactivate HIV latency (*Figure 1 - figure supplement 1*). Remarkably, we found for the first time that the HSV-1 17 strain with ICP34.5 deletion (HSV-ΔICP34.5) could reactivate HIV latency more efficiently than HSV-wt. Specifically, the mRNA levels of HIV genes (LTR, Tat, Gag, Vpr, Vif) were substantially increased in HSV-ΔICP34.5-infected J-Lat 10.6 cells than in HSV-wt treated cells, although there was a weaker replication ability of HSV-ΔICP34.5 in these cells than that of HSV-wt, as indicated by the mRNA level of HSV UL27 (***Figure 1C** and **D***). Furthermore, this finding was verified in ACH-2 cells, derived from T cells latently infected with replication-competent HIV-1. A significantly higher level of p24 protein, a key indicator of HIV replication, was found in the HSV-ΔICP34.5-infected ACH-2 cells than in the HSV-wt treated cells (***Figure 1E***), and the mRNA levels of HIV-1-related genes (LTR, Tat, Gag, Vpr, Vif) were also significantly increased (***Figure 1F***). Subsequently, we generated J-Lat 10.6 cells stably expressing ICP34.5-Flag-Tag (J-Lat 10.6-ICP34.5) using the recombinant lentivirus system and confirmed the expression of the ICP34.5 protein (***Figure 1G***). Our research revealed that HSV-ΔICP34.5 displayed reduced reversal capacity for the latent HIV reservoir in J-Lat 10.6-ICP34.5 cells compared to J-Lat 10.6 cells (***Figure 1G***). Furthermore, in J-Lat 10.6-ICP34.5 cells, the potency of latent reversal agents like phorbol 12-myristate 13-acetate (PMA) and TNF-α was notably reduced when contrasted with J-Lat 10.6 cells (***Figure 1H***). These findings indicate that HSV ICP34.5 can effectively inhibit the reactivation of the HIV latency, and HSV constructs lacking ICP34.5 potentially reactivate HIV latency with high efficiency.

The modified HSV-based constructs effectively reactivated HIV latency by modulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway

Next, the mechanism of potentially reactivating viral latency by the modified HSV-ΔICP34.5-based constructs was explored. J-Lat10.6 cells were infected with HSV-wt and HSV-ΔICP34.5, and our results showed that HSV-ΔICP34.5 significantly enhanced the phosphorylation of IKKα/β, promoted the degradation of IKBα, and thus led to the accumulation of p65 in the nucleus (***Figure 2A***). NF-κB is a well-known host transcription factor that exists in the form of the NF-κB-IκB complex in resting cells, but IκB can degrade and release the NF-κB dimer to enter the nucleus and promote gene transcription in response to external stimulation. Using the coimmunoprecipitation (Co-IP) assay, we verified that the ICP34.5 protein had a specific interaction with IKKα/β, and then ICP34.5 could dephosphorylate IKKα/β. Moreover, the overexpression of ICP34.5 effectively inhibited lipopolysaccharide (LPS) -induced NF-κB pathway activation by inhibiting p65 entry into the nucleus (***Figure 2B-C***).

To further clarify the underlying mechanism, we next performed the IP-MS assay to identify other potential molecules contributing to this reactivation in J-Lat 10.6-ICP34.5 overexpressing cells, and we found that ICP34.5 can also interact with heat shock 1 protein (HSF1) (*table supplement* 1). HSF1 has been reported as a transcription factor correlated with the reactivation of HIV latency (Xu et al., 2022, Lin et al., 2018, Zeng et al., 2017). To test whether HSF1 contributes to the reactivation of HIV latency by HSV-ΔICP34.5-based constructs, KRIBB11, an inhibitor of HSF1, was administered to HSV-ΔICP34.5-infected J-Lat 10.6 cells. The results indicated that the reactivation ability of HSV-ΔICP34.5 was significantly inhibited by KRIBB11 treatment in a dose-dependent manner (***Figure 2D** and **E***). Furthermore, a significant enhancement of the binding of HSF1 to the HIV LTR was observed upon HSV-ΔICP34.5 infection, leading to an increase in the reactivation of HIV latency (***Figure 2F***). The direct interaction between ICP34.5 and HSF1 was also identified by Co-IP assay. Importantly, HSF1 was effectively dephosphorylated at Ser320 as a result of the overexpression of ICP34.5, while no influence on the mRNA or protein expression of HSF1 was observed (***Figure 2, G** and **H***, *Figure 2-figure supplement 2*). Considering that protein phosphatase 1 (PP1) can interact with ICP34.5 and dephosphorylate eIF2α (Li et al., 2011), we

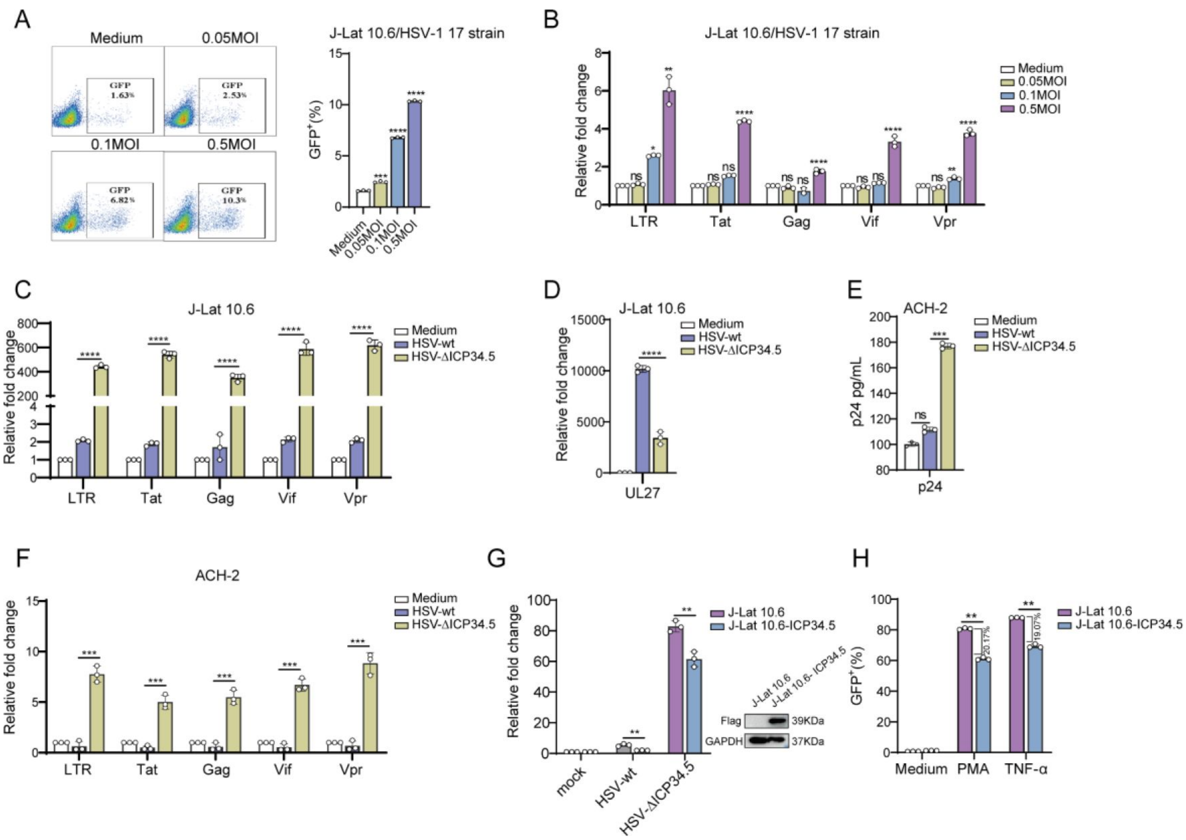

Figure 1.

The modified HSV-ΔICP34.5-based constructs reactivated HIV latency more efficiently than wild-type HSV counterparts.

(A) J-Lat 10.6 cells (1×106) were infected with varying MOIs of wild-type HSV-1 17 strain for 30 h. The increasing expression level of GFP+ cells with increasing MOI of HSV-1 was displayed with the pseudocolor plot (left) and the corresponding bar chart (right). **(B)** The mRNA levels of HIV-1 LTR, Tat, Gag, Vpr, and Vif in J-Lat 10.6 cells following infection with the wild-type HSV-1 17 strain were shown with the histogram. (**C**) J-Lat 10.6 cells ($1{\times}10^6$) were infected with HSV-wt or HSV-ΔICP34.5 at an MOI of 0.1 for 30 h, and then the mRNA levels of HIV-1 LTR, Tat, Gag, Vpr, Vif, and (**D**) HSV-1 UL27 were shown with the histogram. (**E**) ACH-2 cells ($1{\times}10^6$) were infected with HSV-wt or HSV-ΔICP34.5 at an MOI of 0.1 for 30 h, and then the p24 protein level was detected using an HIV-1 p24 ELISA kit. (**F**) ACH-2 cells ($1{\times}10^6$) were infected with HSV-wt or HSV-ΔICP34.5 at an MOI of 0.1 for 30 h, and then the mRNA levels of HIV-1 LTR, Tat, Gag, Vpr and Vif were shown with the histogram. (**G**) J-Lat 10.6 and J-Lat 10.6-ICP34.5 cells were infected with HSV-wt or HSV-ΔICP34.5, and then the mRNA levels of HIV-1 Tat were shown with the histogram (left). The blotting showed that the J-Lat 10.6 cells stably expressing HSV ICP34.5 (J-Lat 10.6-ICP34.5) can appropriately express ICP34.5 protein using Flag-tag antibodies (right). (**H**) J-Lat 10.6 and J-Lat 10.6-ICP34.5 cells were respectively stimulated with PMA (10 ng/mL) and TNF-α (10 ng/mL), and the expression level of GFP^+ cells was displayed with the corresponding bar chart. Data shown are mean ± SD. **P<0.01, ****P*<0.001, *****P*<0.0001. ns: no significance.

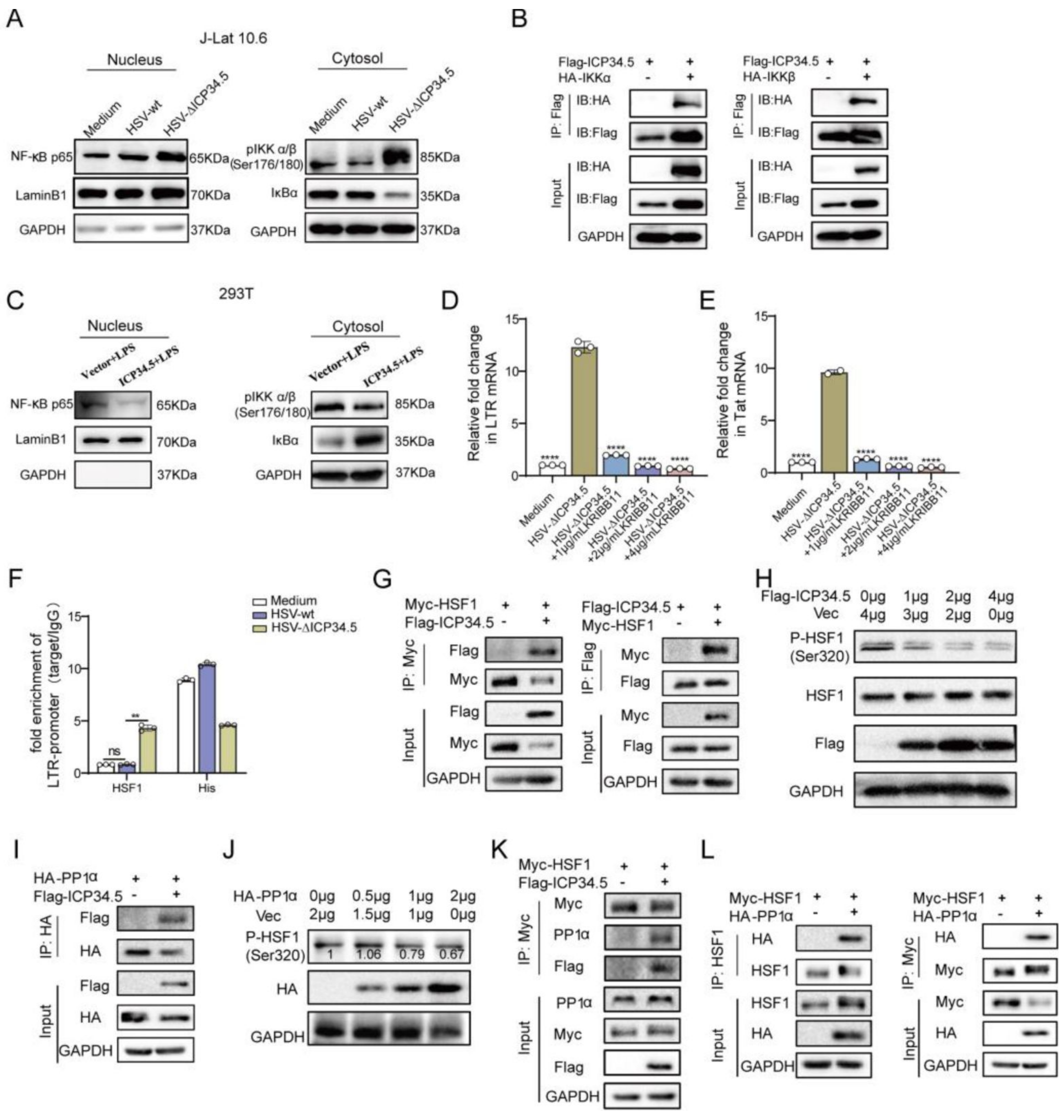

Figure 2.

The modified HSV-based constructs effectively reactivated HIV latency by modulating the NF-κB pathway and HSF1 pathway.

(**A**) J-Lat 10.6 cells were infected with HSV-wt and HSV-ΔICP34.5 at an MOI of 0.1. The cytoplasmic and nuclear proteins were separated to detect the expression level of p65, p-IKKα/β and IkBα. GAPDH and Lamin B1 served as loading controls for cytoplasmic and nuclear proteins, respectively. (**B**) 293T cells were transfected with Flag-ICP34.5, IKKα (left), or IKKβ (right), and the cell lysates and IP complexes were analyzed through Co-IP assays. (**C**) 293T cells were transfected with Flag-ICP34.5 or empty vector (Vec) for 24h, and then treated with LPS (1μg/mL) for 8h. The cytoplasmic and nuclear proteins were analyzed by Western blotting (WB) assay. (**D-E**) J-Lat 10.6 cells were infected with HSV-ΔICP34.5 and then treated with different concentrations of KRIBB11. The relative fold change in LTR and Tat mRNA was analyzed by qPCR. (**F**) J-Lat 10.6 cells were infected with HSV-wt or HSV-ΔICP34.5 at MOI of 0.1 for 36h. ChIP-qPCR was conducted to evaluate the ability of HSF1 to bind to the LTR. IgG and Histone antibody (His) were used as negative and positive controls, respectively. (**G**) 293T cells were transfected with Flag-ICP34.5 and Myc-HSF1, and the cell lysates and IP complexes were analyzed through Co-IP assays. (**H**) 293T cells were transfected with 0, 1, 2, 4 μg of Flag-ICP34.5 and analyzed by WB analysis. (**I**) 293T cells were transfected with Flag-ICP34.5 and HA-PP1α, and the cell lysates and IP complexes were analyzed through Co-IP assays. (**J**) 293T cells were transfected with 0, 0.5, 1, 2 μg of HA-PP1α and analyzed by WB analysis. 293T cells were transfected with Myc-HSF1 with or without Flag-ICP34.5 (**K**), or were transfected with Myc-HSF1 and HA-PP1α (**L**), and then the cell lysates and IP complexes were analyzed through Co-IP assays. Data shown are mean ± SD. ***P*<0.01, *****P*<0.0001. ns: no significance.

then investigated the interaction between PP1α and ICP34.5 (***Figure 2I***). Additionally, a direct interaction between PP1α and HSF1 was found (***Figure 2J-L***), allowing for the dephosphorylation of HSF1 and then affecting its ability to reactivate HIV latency.

Collectively, these findings demonstrated that our modified HSV-ΔICP34.5-based constructs effectively reactivated HIV latency by modulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway.

Construction of recombinant HSV-1 expressing exogenous SIV genes elicited robust immune responses in mice

Prior research suggested that HSV-ΔICP34.5 holds significant promise in reactivating latent HIV efficiently. Subsequently, we explored the potential of advancing HSV-ΔICP34.5 as a bifunctional therapeutic vector to revive latent viral reservoirs and inducing antigen-specific immune responses against chronic HIV infection. To achieve this, the HSV-1 ICP34.5 gene was selectively knocked out, and exogenous antigens were introduced using the bacterial artificial chromosome (BAC)/galactokinase (galK) selection system. Additionally, the ICP47 gene was ablated to augment the immunogenicity of the HSV vector (***Figure 3A***, *Figure 3-figure supplement 3*). A series of recombinant HSV-ΔICP34.5ΔICP47-based vectors expressing SIV gag and env antigen were constructed, and the antigen expression of these constructs was confirmed by Western blotting assay (***Figure 3B** and **C***). Furthermore, to improve the immunogenicity of the targeted antigen, we fused the SIV gag with soluble PD1 (sPD1), enabling it to competitively bind with PD-L1 and thereby block the PD1/PD-L1 immune inhibitory pathway (***Figure 3D***). Consistent with the above findings, these HSV-ΔICP34.5ΔICP47-based SIV vaccines also efficiently reactivated latent HIV proviruses (*Figure 3-figure supplement 4*). Then, the immunogenicity of the above modified HSV-ΔICP34.5ΔICP47-based SIV vaccine was assessed in mice (***Figure 3E***). Our results showed that these constructs effectively elicited SIV antigen-specific T cell immune responses using the interferon γ (IFN-γ) ELISpot assay and the intracellular cytokine staining (ICS) assay (***Figure 3F-M***). Of note, the frequency of SIV Gag-specific IFN-γ -secreting spot-forming cells (SFCs) in the HSV-sPD1-SIVgag group (1350 SFCs per 10^6 splenocytes) was significantly higher than that in the HSV-SIVgag group (498 SFCs per 10^6 splenocytes) (***Figure 3F***). The frequency of SIV Env2-specific IFN-γ -secreting SFCs in the HSV-SIVenv group was significantly higher than the HSV-empty group (***Figure 3G***). Furthermore, the functionality of antigen-specific T cell subsets in response to SIV antigen stimulation was confirmed using the ICS assay (***Figure 3H-M***). Consistently, the HSV-sPD1-SIVgag group showed a significantly higher frequency of SIV-specific $CD3^+$ T, $CD4^+$ T, and $CD8^+$ T cell subsets secreting IFN-γ, IL-2, and TNF-α cytokines compared to the HSV-SIVgag group (***Figure 3I-K***). Notably, a heightened proportion of Gag-specific effector memory T cells (Tem) of $CD8^+$ T cell subset was observed in comparison to the HSV-Gag group (***Figure 3L***). In addition, a higher frequency of SIV-Env2-specific $CD4^+$ T cells secreting IFN-γ was observed in the HSV-SIVenv group than in the HSV-empty group (***Figure 3M***). These data indicated that the vaccines constructed in this study elicit a T cell immune response in mice. Moreover, the blockade of PD1/PDL1 signaling pathway effectively enhanced vaccine-induced T cell immune responses, which was consistent with our and other previous studies (Zhou et al., 2013, Wu et al., 2022, Pan et al., 2018, Xiao et al., 2014).

The modified HSV-based constructs efficiently elicited SIV-specific immune responses in chronically SIV-infected macaques

Next, the immunogenicity of these HSV-vectored SIV vaccines was further investigated in chronically SIVmac239-infected rhesus macaques (RMs). To mimic chronically infected HIV patients in clinic practice, all RMs used in this study were chronically infected with SIV and received ART treatment for several years, as reported in our previous studies (Pan et al., 2018, Yang et al., 2019, Wu et al., 2021, Wu et al., 2022, He et al., 2023). Based on sex, age, viral

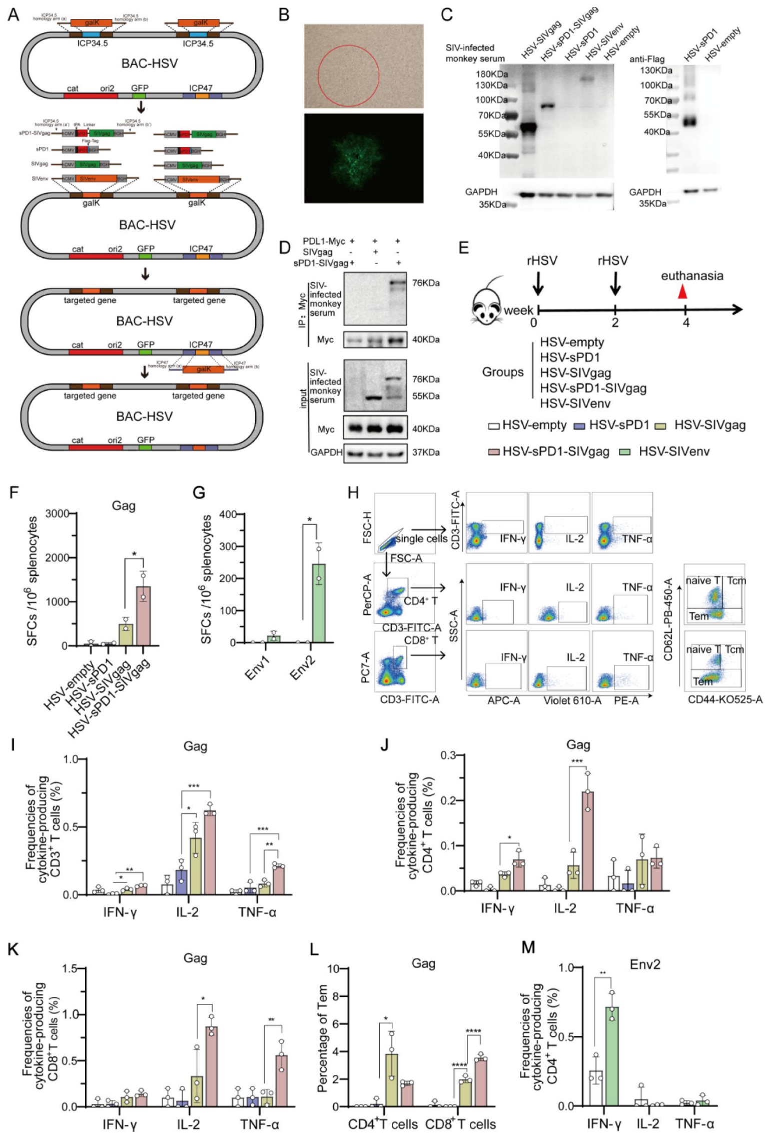

Figure 3.

Recombinant HSV-1 vector-based SIV vaccines induce specific T cell immune responses in mice.

(A) The schematic diagram illustrated the process of constructing recombinant HSV through the BAC/galK selection system. Firstly, ICP34.5 gene was replaced with galK gene through homologous recombination. Subsequently, galK was substituted with the target gene expression cassette containing the hCMV promoter and BGH terminator. Finally, the ICP47 gene was deleted. (B) The image of one clone of the rescued recombinant HSV with the brightfield (top) and the fluorescence channel (bottom). (C) Vero cells were infected with recombinant HSV constructs, and the protein expression of targeted genes was detected by using SIV-infected monkey serum. **(D)** HeLa cells were transfected with Myc-PDL1, followed by infection with HSV-empty, HSV-SIVgag, or HSV-sPD1-SIVgag at an MOI of 0.1 for 24 h. The cell lysates were subjected to Co-IP analysis. (**E**) Schematic schedule of vaccinated mice. Twenty-five mice were randomly allocated to five groups, HSV-empty, HSV-sPD1, HSV-SIVgag, HSV-sPD1-SIVgag, and HSV-SIVenv. At week 0 and week 2, mice were injected with the corresponding vaccines. At week 4, mice were sacrificed and spleen lymphocytes were collected to evaluate immune response. (**F-G**) Column graphs depicted the Gag or Env1, Env2-specific spot-forming cells (SFCs) per 10^6 spleen lymphocytes as measured by IFN-γ ELISpot assay. (**H**) The pseudocolor plot of flow cytometry illustrated the gating strategy. Column graphs depicted the frequencies of IFN-γ, IL-2, and TNF-α production from gag-specific $CD3^+$ T (**I**), $CD4^+$ T (**J**), and $CD8^+$ T cells (**K**). (**L**) The bar chart illustrated the proportion of Tem of $CD4^+$ T and $CD8^+$ T cells under stimulation with the SIV Gag peptide pools. (**M**) The bar chart illustrated the frequencies of Env2-specific IFN-γ^+ $CD4^+$ T cells. These data were expressed as the mean±SD from five mice samples. Three independent experiments for the animal immunization were repeated. **P*<0.05, ***P*<0.01, ****P*<0.001, *****P*<0.0001. ns: no significance.

load, and CD4 count, nine RMs were assigned into three groups: ART+saline group (n=3), ART+HSV-empty group (n=3), and ART+HSV-sPD1-SIVgag/SIVenv group (n=3). All RMs received ART (FTC/6.7 mg/kg/once daily, PMPA/10 mg/kg/once daily) treatment to avoid the interference of free SIV particles. On day 33 and day 52, these RMs were immunized with saline, HSV-empty, or HSV-sPD1-SIVgag/SIVenv respectively. On day 70, ART treatment in all RMs was discontinued to evaluate the time interval of viral rebound. Samples were collected at different time points to monitor virological and immunological parameters (**Figure 4A**, *table supplement* 2). To reduce the impact of individual RM variations, the difference in SIV-specific IFN-γ-secreting SFCs between post-immunization and pre-immunization (ΔSFCs) was used to evaluate the immune response induced by HSV-vectored SIV vaccines. The results showed that SIV Gag-specific ΔSFCs in the ART+HSV-sPD1-SIVgag/SIVenv group were greatly increased when compared with those in the ART+HSV-empty group and ART+saline group (***Figure 4B-D***). A similar enhancement of SIV Gag-specific TNF-α -secreting $CD4^+$ T and $CD8^+$ T subsets was also verified by ICS assay (***Figure 4E***). Collectively, these data demonstrated that the HSV-sPD1-SIVgag/SIVenv construct elicited robust SIV-specific T cell immune responses in ART-treated, SIV-infected RMs.

The modified HSV-based constructs effectively reactivated SIV latency in vivo and delayed viral rebound in chronically SIV-infected, ART-treated macaques

Finally, we investigated the therapeutic efficacy of HSV-sPD1-SIVgag/SIVenv in chronically SIV-infected, ART-treated RMs. Consistent with our previous studies, the plasma viral load (VL) in these RMs was effectively suppressed during ART treatment, but rebounded after ART discontinuation. The VL in the ART+saline group promptly rebounded after ART discontinuation, with an average 8.63-fold increase in the rebounded peak VL compared with the pre-ART VL (***Figure 5A, D** and **E***). However, plasma VL in the ART+HSV-empty group and the ART+HSV-sPD1-SIVgag/SIVenv group exhibited a delayed rebound interval (***Figure 5B-D***). Remarkably, there was a lower rebounded peak VL than pre-ART VL in the ART+HSV-sPD1-SIVgag/SIVenv group (average 12.20-fold decrease), while a higher rebounded peak VL than pre-ART VL in the ART+HSV-empty group (average 2.74-fold increase) (***Figure 5E***). Then, we assessed the potential effect on the latent SIV reservoirs in vivo by administering our modified HSV-based SIV therapeutic constructs in these RMs. Although there were no obvious viral blips observed in these RMs, we found significant suppression of integrated SIV DNA provirus in the ART+HSV-sPD1-SIVgag/SIVenv group. However, the copies of the SIV DNA provirus were significantly improved in the ART+HSV-empty group and ART+saline group (***Figure 5F***). More interestingly, we next assessed the magnitude of SIV Pol antigen-specific immune responses, which could represent to some extent the level of newly generated virions from the reactivated SIV reservoirs, because SIV Pol antigen was not included in our designed vaccine constructs. Specifically, the Pol-specific SFCs amount on Day 70 (511 SFCs/10^6 PBMCs, post-vaccination) was higher than that on Day 33 (315 SFCs/10^6 PBMCs, pre-vaccination) in the ART+HSV-sPD1-SIVgag/SIVenv group. In addition, there was a similar observation in the ART+HSV-empty group. In contrast, the Pol-specific SFCs gradually decreased with the duration of ART treatment in the ART+saline group (***Figure 5G***). In addition, the $CD4^+$ /$CD8^+$ T cell ratio (***Figure 5H***) and body weight (*Figure 5-figure supplement 5*) after treatment were effectively ameliorated in the RMs of the ART+HSV-sPD1-SIVgag/SIVenv group, but not in the ART+HSV-empty group and ART+saline group. Taken together, these findings suggested that the latent SIV reservoirs might be effectively purged because of the effect of simultaneously reactivating viral latency and eliciting SIV-specific immune responses by our modified HSV-based SIV therapeutic constructs, thus resulting in a delayed viral rebound in chronically SIV-infected, ART-treated macaques.

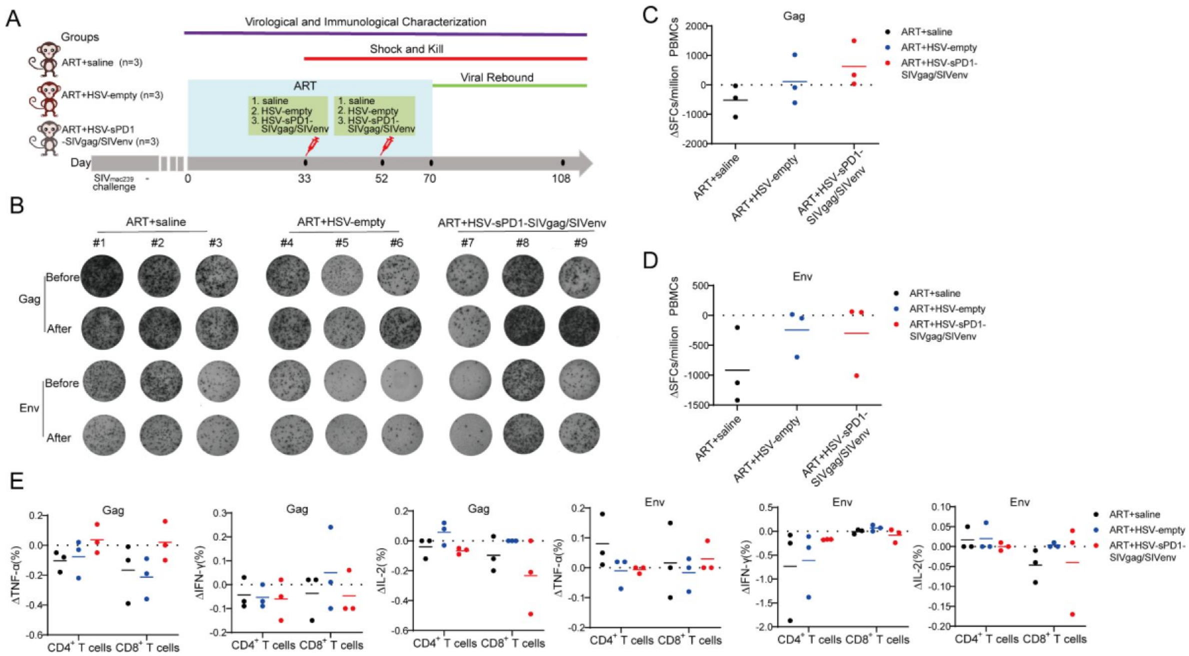

Figure 4.

The modified HSV-based constructs efficiently elicited SIV-specific immune responses in chronically SIV-infected macaques.

(A) Schematic schedule of the macaque experiment. Nine chronically SIV-infected macaques were assigned into three groups: ART+saline group (n=3), ART+HSV-empty group (n=3), and ART+HSV-sPD1-SIVgag/SIVenv group (n=3). All SIV-infected macaques received ART treatment (FTC/6.7 mg/kg/once daily, PMPA/10 mg/kg/once daily) for 33 days. On day 33 and day 52, macaques were immunized with saline, HSV-empty, and HSV-sPD1-SIVgag/SIVenv respectively. On day 70 after the second vaccination, ART treatment was interrupted in all macaques. Samples were collected at different time points to monitor virological and immunological parameters. (B) The representative images for the Gag or Env-specific spots (2.5×105 cells per well) of each macaque pre-vaccination (before, day 33) and post-vaccination (after, day 70) by ELISpot assay. (C-D) The difference in SIV-specific IFN-γ-secreting cells (ΔSFCs) between pre-immunization and post-immunization for the assessment of the immune response induced by HSV-vectored SIV vaccines. (E) The difference in SIV-specific TNF-α/IFN-γ/IL-2-secreting CD4+ T and CD8+ T subsets between pre-immunization and post-immunization was detected by ICS assay.

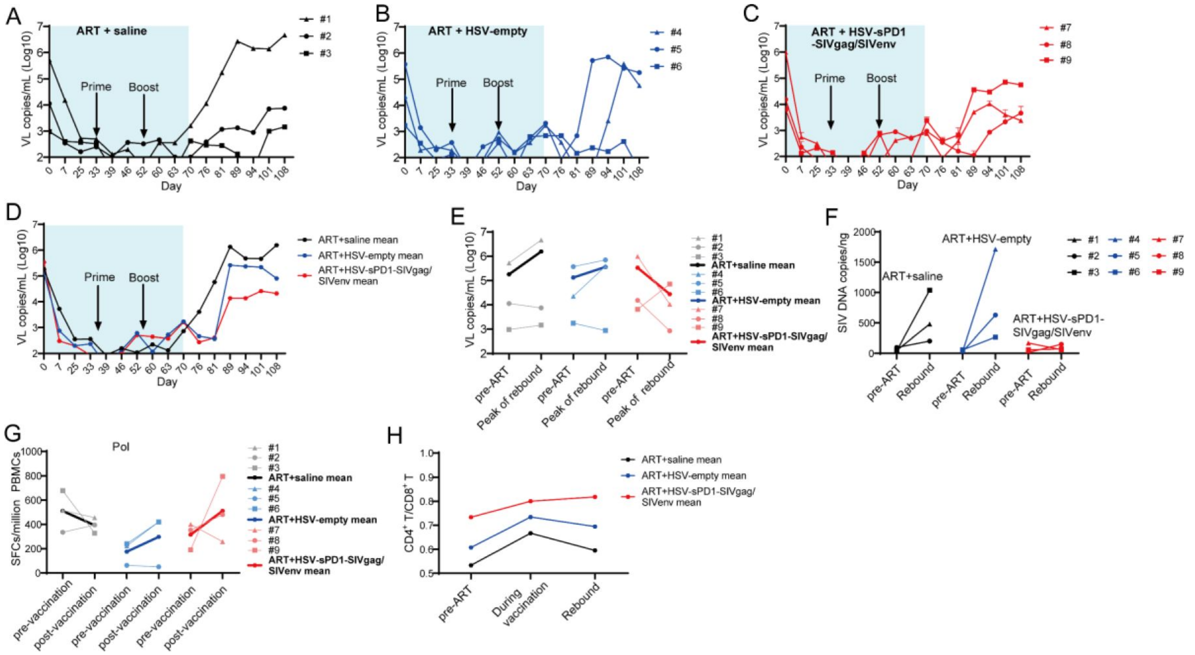

Figure 5.

The modified HSV-based constructs effectively reactivated SIV latency in vivo in chronically SIV-infected, ART-treated macaques.

(A-D) The viral load (VL) change in plasma for each animal was monitored during the whole experiment by y real-time PCR. The detection limit is 100 copies per mL plasma. The shadow represented the duration of ART administration. (E) The VL change in plasma between pre-ART and the peak value in the rebound stage after ART discontinuation. (F) The change of total SIV DNA copies between pre-ART and viral rebound after ART discontinuation. (G) The change of the SIV Pol-specific IFN-γ-secreting cells between pre-immunization (day 33) and post-immunization (day 70) was detected by ELISpot assay. (H) The change of the CD4+ T/ CD8+ T ratio.

Discussion

To conquer the continuous epidemic of AIDS, exploring novel strategies to render and eliminate HIV latency should stand as a pivotal pursuit. Currently, numerous strategies, including shock and kill, block and lock, chimeric antigen receptor T-cell therapy, therapeutic vaccination, and gene editing, have been extensively investigated to target the latent HIV reservoirs for an HIV functional cure (Deeks, 2012, Yeh and Ho, 2021, Maldini et al., 2020, Herzig et al., 2019, Dash et al., 2023, Dashti et al., 2023, Walker-Sperling et al., 2022). However, there is no safe and effective approach for clinical use in HIV-1 patients yet. In the present study, we occasionally found that the modified HSV-ΔICP34.5-based constructs could reactivate HIV latency more efficiently than wild-type HSV counterpart, which inspired us to develop a proof-of-concept strategy based on a bifunctional HSV-vectored therapeutic vaccine, aiming to simultaneously reactivate viral latency and elicit HIV/SIV-specific immune responses for HIV functional cure. Our results indicated that these modified HSV-based constructs efficiently elicited antigen-specific immune responses in mice and chronically SIV-infected macaques, and further therapeutic efficacy experiments showed that this strategy effectively reactivated SIV latency in vivo and delayed viral rebound in chronically SIV-infected, ART-treated macaques (***Figure 6***).

The latent HIV proviruses can harbor it into the host genome with a quiescent transcription state, and thus cannot be recognized by immune surveillance or drug killing (Churchill et al., 2016, Pierson et al., 2000). Therefore, it is critical to disrupt viral latency for developing an HIV cure strategy. Based on our experimental data, the mechanism for efficiently reactivating viral latency by the modified HSV-ΔICP34.5-based constructs may involve regulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway. Indeed, during its replication, HSV can activate the double-stranded RNA-dependent protein kinase (PKR) pathway, and thus phosphorylate the protein translation initiation regulator eIF2α, resulting in the initiation of protein translation (Farassati et al., 2001). Previous studies have shown that the reactivation potential of HSV might be intertwined with NF-κB, Sp1, and other unknown transcription factors by ICP0, ICP4, and ICP27 (Chapman et al., 1991, Amici et al., 2004, Vlach and Pitha, 1992, Mosca et al., 1987b, Golden et al., 1992, Vlach and Pitha, 1993, Mosca et al., 1987a). However, the underlying mechanism by which the ICP34.5-deleted HSV construct can greatly improve the reactivation efficacy of HIV latency remains elusive. ICP34.5 is a neurotoxic factor that can antagonize innate immune responses, including PKR, TANK binding kinase (TBK1) signaling, and Beclin1-mediated apoptosis (He et al., 1997, Manivanh et al., 2017, Orvedahl et al., 2007). ICP34.5 binds to host PP1 and mediates the dephosphorylation of eIF2α, thus allowing protein synthesis and reversing the effects of PKR and host antiviral functions (He et al., 1997, He et al., 1998). In this study, our findings further unveiled an interaction between ICP34.5 and HSF1, resulting in reduced HSF1 phosphorylation via recruitment of PP1α. Interestingly, previous reports indicated that HSF1 could positively regulate HIV gene transcription (Rawat and Mitra, 2011), which is facilitated by its nuclear entry post-phosphorylation and subsequent recruitment of p300 for self-acetylation, along with binding to the HIV-1 LTR. Studies have also shown that HSF1 could further orchestrate p-TEFb recruitment to promote HIV-1 transcriptional elongation (Peng et al., 2020, Lin et al., 2018, Pan et al., 2016b, Pan et al., 2016a). Under stress-induced conditions, phosphorylation triggers the formation of the HSF1 trimer, thus facilitating its nuclear entry to bind to heat shock elements (HSEs) for gene transcription regulation (Bonner et al., 2000, Timmons et al., 2020). Additionally, we also demonstrated that ICP34.5 interacted with IKKα and IKKβ, thereby impeding NF-κB nuclear entry and further curbing HIV latency. Consistently, previous studies have also suggested that ICP34.5 could disrupt the NF-κB pathway and possibly affect the maturation of dendritic cells (Jin et al., 2011). Intriguingly, these findings collectively indicated that ICP34.5 might play an antagonistic role with the reactivation potential of HSV, and thus our modified HSV-ΔICP34.5 constructs reactivate HIV/SIV latency through the release of imprisonment from ICP34.5.

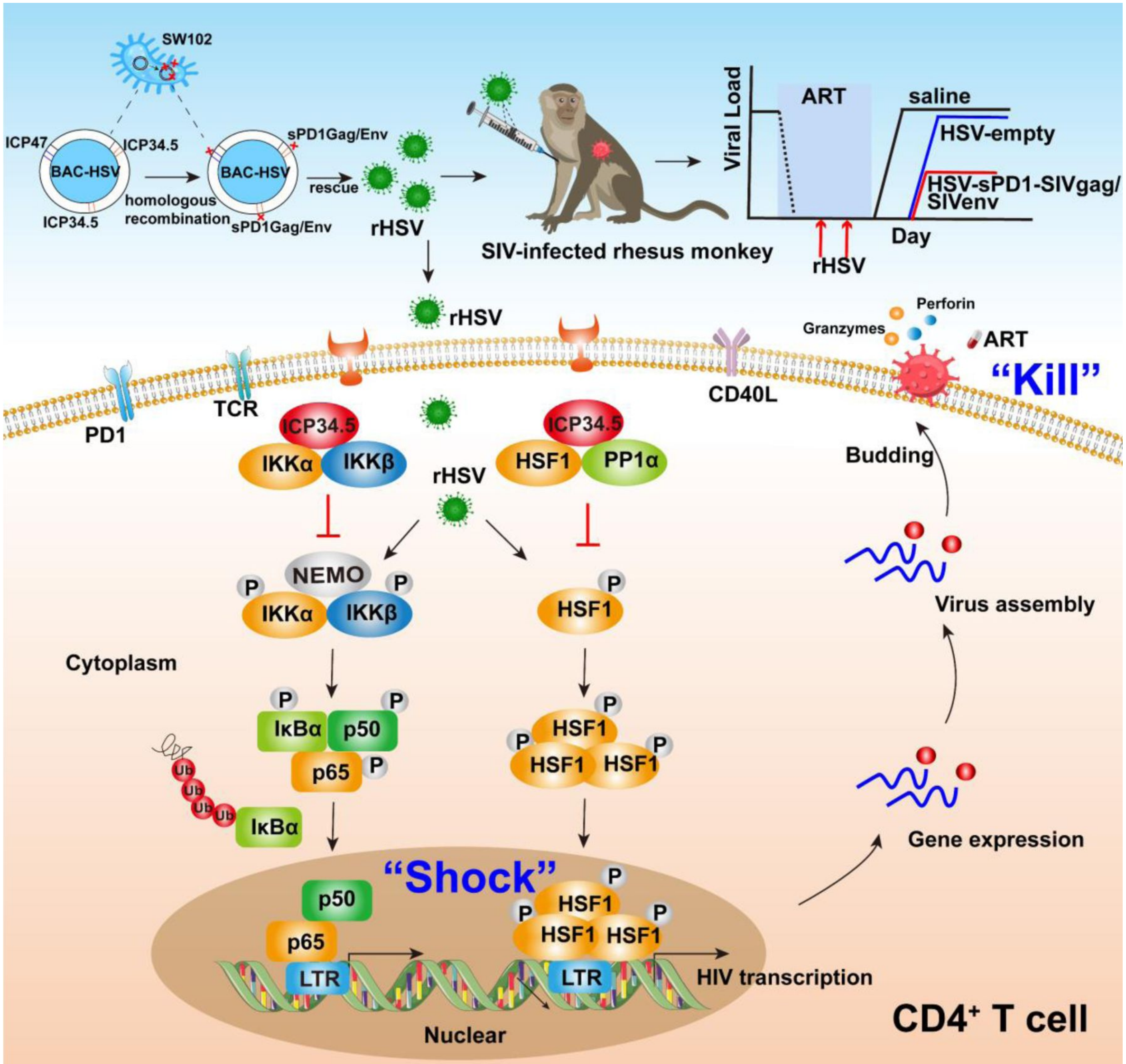

Figure 6.

Pattern to illustrate the proof-of-concept strategy based on a bifunctional HSV-vectored therapeutic vaccine for HIV functional cure.

In the present study, the modified HSV-ΔICP34.5-based constructs effectively reactivated HIV/SIV latency by modulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway (Shock) and simultaneously elicited antigen-specific polyfunctional $CD8^+$ T cells to eliminate cells infected with the reactivated virion (Kill). BAC: bacterial artificial chromosome; rHSV: recombinant HSV; TCR: T-cell receptor; PD1: Programmed Cell Death Protein 1; CD40L: CD40 Ligand.

Another observation in this study is that the HSV-sPD1-SIVgag/SIVenv construct elicited robust and persistent SIV-specific T cell immune responses in ART-treated, chronically SIV-infected macaques. Increasing evidence has indicated that HIV-specific cytolytic T lymphocytes (CTLs) can facilitate the suppression of latent viral reservoirs, and thus, the induction of robust and persistent HIV-specific CTL responses is essential for achieving long-term disease-free and transmission-free HIV control (Collins et al., 2020). Featured polyfunctional CD8$^+$ T cells may contribute to HIV elite controllers or long-term non-progressors, a rare proportion of HIV-infected individuals who can spontaneously control viral replication even without ART treatment (Owen et al., 2010, Ferre et al., 2009, O'Connell et al., 2009). In addition, both our previous study and others have demonstrated that strong antigen-specific CD8$^+$ T cell immune responses, especially effector memory CD8$^+$ T cells, were associated with a lower viral load, and in vivo CD8$^+$ lymphocyte depletion with intravenous infusion of anti-CD8 monoclonal antibody could lead to dramatical viral rebound in these vaccinated elite macaques (Perdomo-Celis et al., 2022, Pandrea et al., 2011). Notably, previous studies have shown that HSV-based vaccines expressing HIV/SIV antigen did elicit specific CD8$^+$ T cell immune responses in mice and macaques, but the magnitude was not as strong as other viral vectors (Parker et al., 2007, Murphy et al., 2000, Kaur et al., 2007). Therefore, to further improve its immunogenicity in vivo, some modifications were adopted to optimize the HSV-vectored vaccine. 1) The ICP47 protein, encoded by the US12 gene, can bind to the transporter associated with antigen presentation (TAP) 1/2, inhibiting the transport of viral peptides into the endoplasmic reticulum and the loading of peptides onto nascent major histocompatibility complex (MHC) class I molecules to activate CD8$^+$ T cells (Orr et al., 2005). Therefore, the ICP47 gene was deleted from our developed HSV vector. 2) Negative immunoregulatory molecules, including PD1, TIM-3, and LAG-3, are usually involved in the pathogenesis of HIV infection, as well as in chronically SIV-infected macaques. Among them, PD1 upregulation can result in the exhaustion and dysfunction of CD8$^+$ T cells (Trautmann et al., 2006). In addition, PD1 expression on memory CD4$^+$ T cells might be linked with HIV latency. In this study, we modified the SIV antigen by fusing to soluble PD1 (sPD1), which can block the PD1/PDL1 pathway by competitively binding with PDL1 and thus improve HIV/SIV vaccine-induced CD8$^+$ T cell immune responses (Zhou et al., 2013, Wu et al., 2022, Pan et al., 2018, Xiao et al., 2014).

Although promising, there are still some limitations to our study. First, this is a pilot study with relatively small numbers of RMs, and future studies with a larger number of animals can be conducted to better verify our strategy. Second, the HSV-sPD1-SIVgag/SIVenv vaccine resulted in delayed viral rebound and a lower peak viral load post-rebound than pre-ART treatment but did not completely suppress SIV virus rebound in this study, which may be attributed to suboptimal doses and treatment, implying that we should further optimize this regimen for eventually achieving an HIV functional cure in future studies. Taken together, these findings demonstrated that our modified HSV-ΔICP34.5-based constructs potentially reactivated HIV/SIV latency by modulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway, and thereby we developed a proof-of-concept HIV functional cure strategy based on a bifunctional HSV-vectored therapeutic vaccine, which can provide insights into the rational design of novel strategies for pursuing an HIV functional cure.

Methods

Mouse ethics statement and vaccination

Female BALB/c mice, aged six to eight weeks, were procured from the Experimental Animal Center of Sun Yat-sen University. A total of twenty-five mice were randomly divided into five groups to evaluate the immunogenicity of the recombinant HSV-vector-based SIV vaccine. During the initial week, each mouse was subcutaneously administered a vaccination of $1{\times}10^6$ PFU of the respective recombinant HSV-vector vaccines (HSV-empty, HSV-sPD1, HSV-SIVgag, HSV-sPD1-SIVgag, HSV-

SIVenv). Following a two-week interval, a booster vaccination was administered via the subcutaneous route using 2×10^6 PFU of recombinant HSV-vector vaccines. The subsequent assessment of the immune response involved ELISpot and intracellular cytokine staining (ICS) assays in accordance with the vaccination schedule.

Ethics statement and vaccination of macaques

A total of 9 chronically SIVmac239-infected rhesus macaques (RMs) were included in this study. All RMs received ART treatment (FTC/6.7 mg/kg/once daily, PMPA/10 mg/kg/once daily) for a duration of 33 days. The RMs were allocated into three groups based on their baseline plasma SIV viral loads (VL): ART + saline (n=3) as the control group, ART+ HSV-empty (n=3) as the sham group, and ART+ HSV-sPD1-SIVgag/SIVenv (n=3) as the vaccinated group. On day 0, all RMs received ART treatment. Once the VL dropped below the limit of detection (1×10^2 copies/mL), the vaccinated group and sham group were primed with a subcutaneous vaccination of 1×10^7 PFU HSV-sPD1-SIVgag/SIVenv or HSV-empty, respectively, while the control group received an injection of 0.9% saline. After three weeks of the prime vaccination, the vaccinated group and sham group RMs received a subcutaneous booster vaccination of 5×10^7 PFU HSV-sPD1-SIVgag/SIVenv or HSV-empty, respectively. After two weeks of booster vaccination, the ART treatment for all RMs was interrupted. The immune response was evaluated by IFN-γ ELISpot and ICS assays. Plasma viral load was regularly monitored throughout the studied period.

Peptide pools

The peptide pools, encompassing the complete sequences of SIV Gag, Env, and Pol proteins, comprising 15 amino acids with 11 overlaps, were generously provided by the HIV Reagent Program, National Institutes of Health (NIH), USA. Gag pools comprise 125 peptides, while Env and Pol pools are subdivided into Env1 (109 peptides) and Env2 (109 peptides) pools, as well as Pol1 (131 peptides) and Pol2 (132 peptides) pools. These peptide pools were dissolved in dimethyl sulfoxide (DMSO, Sigma) at a concentration of 0.4 mg/ peptide/mL for subsequent immunological assays.

Construction of recombinant HSV

For the generation of recombinant HSV-vector-based vaccines, we implemented modifications at the ICP34.5 loci through homologous recombination within the BAC-HSV-1 system. Specifically, double copies of the ICP34.5 and ICP47 genes were either deleted or inserted into the respective counterparts: sPD1, SIVgag, sPD1-SIVgag, and SIVenv genes. The sPD1-SIVgag gene was created by fusing the N-terminal region of mouse soluble PD1 (sPD1) with the C-terminal segment of SIVgag, connected by a GGGSGGG linker, which was engineered through overlapping PCR.

IFN-γ ELISpot assay

The IFN-γ-ELISpot assay was conducted in accordance with our previous study (Sun et al., 2010). In the mouse experiment, 2.5×10^5 freshly isolated mouse spleen lymphocytes were simulated with SIV Gag, Env1, and Env2 peptide pools. In the monkey experiment, 2×10^5 peripheral blood mononuclear cells (PBMCs) were seeded and simulated with the SIV Gag, Env, and Pol peptide pools. Mock stimulation utilized DMSO (Sigma), while concanavalin A (ConA, 10 μg/mL) was employed as a positive control. Spot quantification was performed using an ELISpot reader (Mabtech), and peptide-specific spot counts were determined by subtracting the spots from mock stimulation.

Intracellular cytokine staining (ICS)

The ICS assay was performed as described previously (Wu et al., 2022). In the mouse experiment, 2×10^6 freshly isolated mouse spleen lymphocytes were stimulated with SIV Gag, Env1, and Env2 peptide pools. For the monkey experiment, 2×10^6 PBMCs were seeded and simulated with the SIV Gag, Env, and Pol peptide pools. DMSO and PMA (Thermo Scientific) plus ionomycin were used as the mock and positive controls, respectively. Data analysis was performed using FlowJo software (version 10.8.1), and the antibodies used are listed in *table supplement* 3. The frequencies of cytokines produced from peptide-specific cells were analyzed by subtracting mock stimulation.

SIV viral RNA and DNA copy assays and absolute T cell count

Absolute T cell count and the levels of plasma mRNA and SIV total DNA were quantified as described previously (Wu et al., 2022). Plasma viral RNA copy numbers were determined via SYBR green-based real-time quantitative PCR (Takara), using SIV gag-specific primers (*table supplement 4*). Viral RNA copy numbers were calculated based on the standard curve established using SIVmac239 gag standards. The lower limit of detection for this assay was 100 copies/mL of plasma. Total cellular DNA was extracted from approximately 0.5 to 5 million cells using a QIAamp DNA Blood Minikit (Qiagen). PCR assays were performed with 200 ng samples of DNA, and SIV viral DNA was quantified using a pair of primers targeting a conserved region of the SIV gag gene, as previously described. Quantitation was performed by comparing the results to the standard curve of SIV gag copies.

Plasmids, cells, and viruses

pVAX-Myc-PDL1, pVAX-Flag-ICP34.5, pVAX-Myc-HSF1, pVAX-HA-PP1α: full-length mouse PDL1, full-length HSV-1 ICP34.5, and full-length human HSF1 and human PP1α with N or C-terminal Tag were cloned into the pVAX vector. pVAX-empty, pcDNA3.1-IKKα, and pcDNA3.1-IKKβ plasmids were stored in our laboratory. pVAX-empty was used as a mock transfection in our study.

293T cells (from the embryonic kidney of a female human fetus), Vero cells (from the kidney of a female normal adult African green monkey), and Hela cells (from the cervical cancer cells of an African American woman) were cultured in complete Dulbecco's modified Eagle's medium (DMEM, Gibco) containing 10% fetal bovine serum (FBS, Gibco) and 1% penicillin/streptomycin (Gibco) at 37°C in an atmosphere of 5% CO2. The J-Lat 10.6 cells (Jurkat cells contain the HIV-1 full-length genome whose Env was frameshifted and inserted with GFP in place of Nef) and the HIV-1 latently infected $CD4^+$ CEM cells ACH-2 (A3.01 cell integrated HIV-1 proviral DNA) were cultured in complete RPIM640 (Gibco) containing 10% FBS and 1% penicillin/streptomycin at 37°C in an atmosphere of 5% CO2. J-Lat 10.6-ICP34.5 cells were constructed in our laboratory.

The HSV-1 (McKrae) and HSV-1 (F) strains were stored in our laboratory. HSV-1 (17 strain) was rescued from pBAC-GFP-HSV in Vero cells. HSV-ΔICP34.5 was rescued from pBAC-GFP-HSV-ΔICP34.5 (with double copies of ICP34.5 deleted) in Vero cells.

RT-qPCR

RT-qPCR was performed as described in our previous study (Zhao et al., 2022). Data were normalized to β-actin. The primer sequences are listed in Supplemental Table 4. Fold changes in the threshold cycle (Ct) values were calculated using the $2^{-\Delta\Delta Ct}$ method.

Chromatin immunoprecipitation (ChIP) assay

ChIP analysis was conducted using the SimpleChIP Enzymatic Chromatin IP kit (Agarose Beads) (CST), following the manufacturer's instructions. Quantitative real-time PCR was employed for detecting the LTR sequence. The sequences of primers used in the LTR ChIP are listed in Supplemental Table 4. % Input= 2% × $_2$[(ct)2% input sample (ct) IP sample].

Co-immunoprecipitation (Co-IP)

Cells were harvested and subjected to Co-IP assay, following the protocol outlined in our previous study (Zhao et al., 2022).

Western blotting analysis

Nuclear and cytoplasmic proteins were extracted by kits following the manufacturer's instructions (Beyotime). The Western blotting assay was performed as previously described (Zhao et al., 2022).

Statistical analysis

GraphPad Prism 8.0 (GraphPad Software, San Diego, California) was used for statistical analysis. For intragroup direct comparisons, Student's unpaired two-tailed *t* test was performed to analyze significant differences. For comparisons of multiple groups, one-way ANOVAs were performed. Significance levels are indicated as $*P < 0.05$, $** P < 0.01$, $*** P < 0.001$, $**** P < 0.0001$.

Study approval

Mice experiment was approved by the Laboratory Animal Ethics Committee guidelines at Sun Yat-sen University (approval number: SYSU-IACUC-2021-000185). Chinese rhesus macaques (Macaca mulatta) were housed at the Landau Animal Experimental Center, Guangdong Landau Biotechnology Co., Ltd. (approval number: N2021101).

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Additional information

Competing interests

The authors have declared that no conflict of interest exists.

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Author Contributions

All authors were involved in drafting or critically revising the manuscript for important intellectual content, and all authors approved the final version for publication. CS conceived the project and provided the main funding. CS and LC designed the experiments and reviewed the manuscript. ZW, PL, YY, and CW performed most of the experiments and analyzed the data. ZW, PL, YY, CW, ML, HW, MS, YH, and MC established the animal models and performed animal experiments. ZW and CS interpreted the data and wrote the manuscript.

Additional files

Supplementary files

Figure S1-S5. Table S1-S4.

Data Availability Statement

Source data contain the numerical data used to generate the figures.

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Reviewer #1 (Public Review):

Summary:

The authors constructed a novel HSV-based therapeutic vaccine to cure SIV in a primate model. The novel HSV vector is deleted for ICP34.5. Evidence is given that this protein blocks HIV reactivation by interference with the NFkappaB pathway. The deleted construct supposedly would reactivate SIV from latency. The SIV genes carried by the vector ought to elicit a strong immune response. Together the HSV vector would elicit a shock and kill effect. This is tested in a primate model.

Strengths and weaknesses:

(1) Deleting ICP34.5 from the HSV construct has a very strong effect on HIV reactivation. Why is no eGFP readout given in Figure 1C as for WT HSV? The mechanism underlying increased activation by deleting ICP34.5 is only partially explored. Overexpression of ICP34.5 has a much smaller effect (reduction in reactivation) than deletion of ICP34.5 (strong activation); so the story seems incomplete.

(2) No toxicity data are given for deleting ICP34.5. How specific is the effect for HIV reactivation? An RNA seq analysis is required to show the effect on cellular genes.

(3) The primate groups are too small and the results to variable to make averages. In Figure 5, the group with ART and saline has two slow rebounders. It is not correct to average those with a single quick rebounder. Here the interpretation is NOT supported by the data.

Discussion

HSV vectors are mainly used in cancer treatment partially due to induced inflammation. Whether these are suitable to cure PLWH without major symptoms is a bit questionable to me and should at least be argued for.

https://doi.org/10.7554/eLife.95964.1.sa1

Reviewer #2 (Public Review):

Summary:

In this article, Wen et. al. describe the development of a 'proof-of-concept' bi-functional vector based on HSV-deltaICP-34.5's ability to purge latent HIV-1 and SIV genomes from cells. They show that co-infection of latent J-lat T-cell lines with an HSV-deltaICP-34.5 vector can reactivate HIV-1 from a latent state. Over- or stable expression of ICP 34.5 ORF in these cells can arrest latent HIV-1 genomes from transcription, even in the presence of latency reversal agents. ICP34.5 can co-IP with- and de-phosphorylate IKKa/b to block its interaction with NF-k/B transcription factor. Additionally, ICP34.5 can interact with HSF1 which was identified by mass-spec. Thus, the authors propose that the latency reversal effect of HSV-deltaICP-34.5 in co-infected JLat cells is due to modulatory effects on the IKKa/b-NF-kB and PP1-HSF-1 pathway.

Next, the authors cleverly construct a bifunctional HSV-based vector with deleted ICP34.5 and 47 ORFs to purge latency and avoid immunological refluxes, and additionally, expand the application of this construct as a vaccine by introducing SIV genes. They use this 'vaccine' in mouse models and show the expected SIV-immune responses. Experiments in rhesus macaques (RM), further elicit the potential for their approach to reactivate SIV genomes and at the same time block their replication by antibodies. What was interesting in the SIV experiments is that the dual-functional vector vaccine containing sPD1- and SIV Gag/Env ORFs effectively delayed SIV rebound in RMs and in some cases almost neutralized viral DNA copy detection in serum. Very promising indeed, however, there are some questions I wish the authors had explored to get answers to, detailed below.

Overall, this is an elegant and timely work demonstrating the feasibility of reducing virus rebound in animals, with the potential to expand to clinical studies. The work was well-written, and sections were clearly discussed.

Strengths:

The work is well designed, rationale explained, and written very clearly for lay readers.

Claims are adequately supported by evidence and well-designed experiments including controls.

Weaknesses:

(1) While the mechanism of ICP34.5 interaction and modulation of the NF-kB and HSF1 pathways are shown, this only proves ICP34.5 interactions but does not give away the mechanism of how the HSV-deltaICP-34.5 vector purges HIV-1 latency. What other components of the vector are required for latency reversal? Perhaps serial deletion experiments of the other ORFs in the HSV-deltaICP-34.5 vector might be revealing.

(2) The efficacy of the HSV vaccine vectors was evaluated in Rhesus Macaque model animals. Animals were chronically infected with SIV (a parent of HIV), treated with ART, challenged with bi-functional HSV vaccine or controls, and discontinued treatment, and the resulting virus burden and immune responses were monitored. The animals showed SIV Gag and Env-specific immune responses, and delayed virus rebound (however rebound is still there), and below-detection viral DNA copies. What would make a more convincing argument to this reviewer will be data to demonstrate that after the bi-functional vaccine, the animals show overall reduction in the number of circulating latent cells. The feasibility of obtaining such a result is not clearly demonstrated.

(3) The authors state that the reduced virus rebound detected following bi-functional vaccine delivery is due to latent genomes becoming activated and steady-state neutralization of these viruses by antibody response. This needs to be demonstrated. Perhaps cell-culture experiments from specimens taken from animals might help address this issue. In lab cultures one could create environments without antibody responses, under these conditions one would expect a higher level of viral loads to be released in response to the vaccine in question.

(4) How do the authors imagine neutralizing HIV-1 envelope epitopes by a similar strategy? A discussion of this point may also help.

(5) I thought the empty HSV-vector control also elicited somewhat delayed kinetics in virus rebound and neutralization, can the authors comment on why this is the case?

https://doi.org/10.7554/eLife.95964.1.sa0

Author response:

> ***Reviewer #1 (Public Review):***
>
> *(1) Deleting ICP34.5 from the HSV construct has a very strong effect on HIV reactivation. Why is no eGFP readout given in Figure 1C as for WT HSV? The mechanism underlying increased activation by deleting ICP34.5 is only partially explored. Overexpression of ICP34.5 has a much smaller effect (reduction in reactivation) than deletion of ICP34.5 (strong activation); so the story seems incomplete.*

Thank you for your comment. (1) In Figure 1c, "HSV-wt" refers to the virus rescued from pBAC—GFP-HSV (as mentioned in the “Method” section), which carries GFP itself. Therefore, detecting GFP cannot distinguish between HSV infection and HIV reactivation. Hence, we assess the reactivation effect by measuring the mRNA levels of HIV LTR. (2) Our data indicate that overexpression of ICP34.5 inhibits the reactivation of the HIV latent reservoir, but this effect is not equivalent to the activation observed in HSV-1 with ICP34.5 deletion. There are some possible reasons: one is that the overexpression of ICP34.5 by lentivirus is randomly integrated into the genome of J-Lat cell line, which will potentially activate HIV latency to some extent. The other is that ICP34.5 mainly inhibited HIV reactivation through modulation of host NF-κB or HSF1 pathways, while PMA, TNF-a, and HSV-1 with deleted ICP34.5 can reactivate HIV latency by other mechanisms that have yet to be determined. Thereby, exerting a synergistic small inhibitory effect. We will further discuss this issue in the revised version. Thank you.

> *(2) No toxicity data are given for deleting ICP34.5. How specific is the effect for HIV reactivation? An RNA seq analysis is required to show the effect on cellular genes.*

Thank you for your comment. We plan to conduct several experiments to demonstrate a reduction in HSV-1 replication after ICP34.5 deletion: (1) Detect the growth curve of HSV-1 deleted with ICP34.5 in Vero cells. The virus growth curve of HSV-1 with deleted ICP34.5 may be lower than that of wild-type HSV-1, which could demonstrate a reduction in HSV-1 replication after ICP34.5 deletion. (2) Detect the level of inflammatory factors in tumor cells after infection with HSV-1 deleted with ICP34.5.

We believe that the effect is specific, as we previously tested poxviruses and adenoviruses and found no activation of the latent reservoir. We consider the activation observed with HSV-1 virus and HSV-1 with deleted ICP34.5 to be specific. We will supplement relevant data in the revised version.

In addition, we will provide the corresponding RNA-seq data to assess its effect on cellular genes.

> *(3) The primate groups are too small and the results to variable to make averages. In Figure 5, the group with ART and saline has two slow rebounders. It is not correct to average those with a single quick rebounder. Here the interpretation is NOT supported by the data.*

We agree with you that this is a pilot study of limited numbers of rhesus macaques. There were only 3 monkeys per group in this study, but our results were encouraging. Although the number of macaques was relatively limited, these nine macaques were distributed very carefully based on age, sex, weight and genotype. All SIV-infected macaques used in this study had a long history of SIV infection and had several courses of ART therapy, which mimics treatment of chronic HIV-1 infection in humans. These macaques were infected with SIVmac239 for more than 5 years, and highly pathogenic SIV-infected macaques have been well-validated as a stringent model to recapitulate HIV-1 pathogenesis and persistence during ART therapy in humans. Indeed, in our rhesus model, ART treatment effectively suppressed SIV infection to undetectable levels in plasma, and upon ART discontinuation, virus rapidly rebounded, which is very similar with that in ART-treated HIV patients. Our further studies will be expanded the scale of animals and then to preclinical and clinical study in our next projects. Thank you for your understanding.

> *Discussion*
>
> *HSV vectors are mainly used in cancer treatment partially due to induced inflammation. Whether these are suitable to cure PLWH without major symptoms is a bit questionable to me and should at least be argued for.*

We will provide more data about the safety assessment of HSV-1 vector in SIV-infected macaques, and also further discuss the potential of inflammatory HSV vector in PLWH in the revised manuscript.

> ***Reviewer #2 (Public Review):***
>
> *(1) While the mechanism of ICP34.5 interaction and modulation of the NF-kB and HSF1 pathways are shown, this only proves ICP34.5 interactions but does not give away the mechanism of how the HSV-deltaICP-34.5 vector purges HIV-1 latency. What other components of the vector are required for latency reversal? Perhaps serial deletion experiments of the other ORFs in the HSV-deltaICP-34.5 vector might be revealing.*

We agree with your suggestion. In fact, we are currently further exploring some viral genes of HSV-1 that play a role in activation. We have found that the ICP0 gene of HSV-1 virus can activate HIV, and the specific mechanism is under investigation.

> *(2) The efficacy of the HSV vaccine vectors was evaluated in Rhesus Macaque model animals. Animals were chronically infected with SIV (a parent of HIV), treated with ART, challenged with bi-functional HSV vaccine or controls, and discontinued treatment, and the resulting virus burden and immune responses were monitored. The animals showed SIV Gag and Env-specific immune responses, and delayed virus rebound (however rebound is still there), and below-detection viral DNA copies. What would make a more convincing argument to this reviewer will be data to demonstrate that after the bi-functional vaccine, the animals show overall reduction in the number of circulating latent cells. The feasibility of obtaining such a result is not clearly demonstrated.*

Thank you for your suggestion. We will plan to conduct IPDA experiments to further supplement data on the overall reduction in circulating latent cell numbers in animals.

> *(3) The authors state that the reduced virus rebound detected following bi-functional vaccine delivery is due to latent genomes becoming activated and steady-state neutralization of these viruses by antibody response. This needs to be demonstrated. Perhaps cell-culture experiments from specimens taken from animals might help address this issue. In lab cultures one could create environments without antibody responses,*

> *under these conditions one would expect a higher level of viral loads to be released in response to the vaccine in question.*

We plan to use primary cells for related experiments to further validate the results of the cell experiments.

> *(4) How do the authors imagine neutralizing HIV-1 envelope epitopes by a similar strategy? A discussion of this point may also help.*

Thank you for your comments. In fact, our study adopts the "shock and kill" strategy, with a focus on the "kill" aspect leaning towards T-cell therapy. Although the vaccine in the paper also utilizes Env antigen, we believe these antibodies are insufficient for neutralizing the mutated SIV virus. We strongly agree with your suggestion that in HIV/AIDS treatment, effective T-cell killing combined with broad-spectrum neutralizing antibodies would be more effective. This aligns with our findings, as our treatment has partially delayed viral rebound but with a relatively short duration of suppression. This may indicate insufficient killing activity. In future research, we will further consider the role of broad-spectrum neutralizing antibodies. Our revised manuscript will elaborate on this in the discussion section.

> *(5) I thought the empty HSV-vector control also elicited somewhat delayed kinetics in virus rebound and neutralization, can the authors comment on why this is the case?*

We agree with you that the HSV-1 empty vector does exhibit somewhat a delayed rebound. The reason is that our treatment simultaneously utilizes both the HSV vector vaccine and ART therapy. Although the empty HSV-vector cannot elicit SIV-specific CTL response, it effectively activates the latent SIV reservoirs and then these activated virions can be partially killed by ART, Therefore, even without carrying antigens, the slight delay may be achieved.